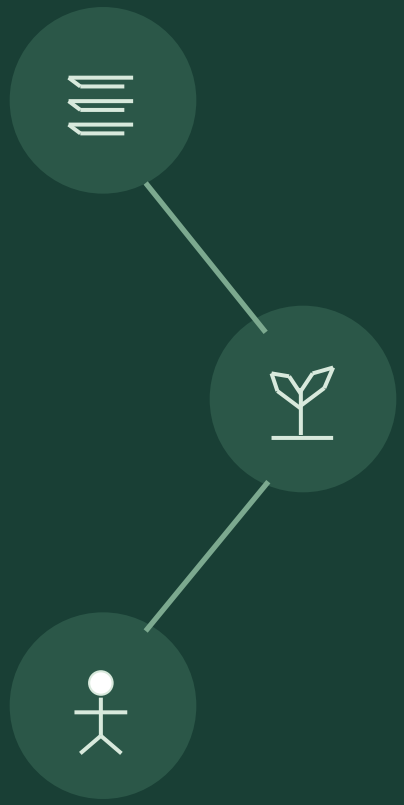
Reasoning with Food Knowledge Graphs



Evidence-based pathways connecting soil, crop health and human health

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Agriculture and health evidence is fragmented

A relationship may be described across several papers, with different entities, conditions and outcomes.



Soil and farming

Soil properties
Management practices

Crop health

Diseases and pathogens
Crop responses

Food composition

Nutrients and toxins
Food quality

Human health

Exposure routes
Biological outcomes

Research need Connect claims while retaining what each paper actually supports.

Research objectives and domain scope

Build a cumulative graph that researchers can inspect from a pathway down to its exact evidence.

1 **Represent scientific relationships**

Extract direct biological and food-quality claims with their conditions.

2 **Connect evidence across papers**

Align shared entities and assemble candidate pathways.

3 **Assess supporting and opposing findings**

Record external evidence separately for every claim.

4 **Enable research inspection**

Explore the graph, original passages and evidence decisions together.

Each graph edge retains its scientific context

A triplet identifies the relationship. Conditions and source evidence determine how to interpret it.

Infected wheat grains → may contain → **Mycotoxins**

ENTITY ROLES

Crop material / contaminant

BIOLOGICAL CONTEXT

Fusarium head blight infection

CERTAINTY

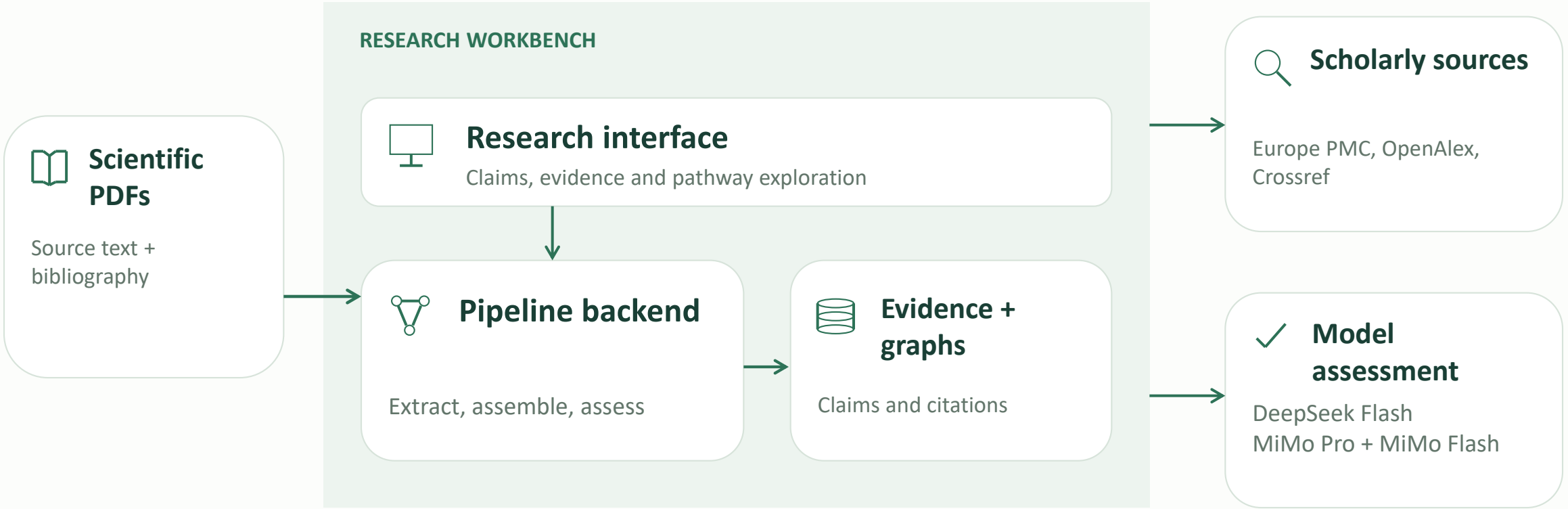
Conditional: “may”

PROVENANCE

Paper, page, sentences and citations

System architecture

The workbench links scientific documents, evidence assessment and an inspectable knowledge graph.



Source documents and external evidence remain traceable throughout the workflow.

Current trust-graph pipeline · end-to-end evidence and model flow

1 GROUND THE CLAIMS IN THE PDF

Assembly comes before external validation

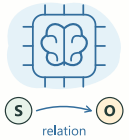


Original research PDF
Pages, bibliography, citations



Structure & number the text
pypdf · chapter-aware references
Target sentences + reading context

L1 · DeepSeek v4.1 Flash



Extract atomic claims
Triplets + conditions + sentence IDs
800-word target + surrounding context

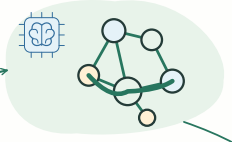
L2 · DeepSeek v4.1 Flash



Check source fidelity
Exact PDF context, direction & certainty
Per-claim verdicts · batches of 3 by default

Entailed claims proceed

L3 · MiMo v2.6 Flash



Assemble provisional paths
LLM aligns entity identities
Code connects the source-grounded edges



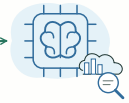
Related claims + standalone claims
Bounded groups

each claim keeps its own evidence, conditions and decision

2 RETRIEVE THE RESEARCH CONTEXT

RAG selection finds evidence; ranking is not a scientific verdict

L4 · MiMo v2.6 Flash



Plan literature queries
Full claim + essential conditions
Independent plan for every claim
API batches default to 6 claims



Independent discovery
Europe PMC + OpenAlex

Source references



Attached PDF citations
Bibliography → DOI / Crossref



Rank article candidates
BGE embeddings + BM25
MedCPT cross-encoder reranking
Identity dedupe · source paper excluded



Fetch & prepare papers
XML / public PDF / indexed abstract
Chunks + exact numbered sentences



Rerank exact passages
BGE + lexical score → MedCPT
Full claim + neutral endpoint query
Default 4 chunks per paper



Shared paper & passage index
SQLite · exact text · sentence IDs · reusable embeddings

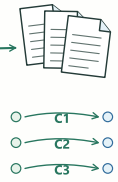
Up to 8 discovered papers / claim by default, plus all usable resolved source citations

selected, identified paper packets

3 JUDGE EACH PAPER, THEN SYNTHESIZE THE GROUP

A separate verdict and proof for every claim

Shared claim group



Same fixed claims;
paper identity stays attached

Selected paper evidence



Paper A



Paper B



Paper … N

repeat for every selected paper

L5 · DeepSeek v4.1 Flash
1 paper + related claims / call







Paper-bound findings
IDs + literal sentence checks

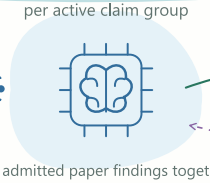






verdict · conditions · exact proof

L6 · MiMo v2.6 Pro
ONE cross-paper synthesis
per active claim group



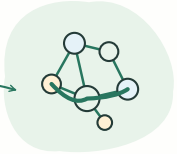
All admitted paper findings together
Separate support / conflict per edge
Scope, types & adjacent-link compatibility



DeBERTa NLI
Diagnostic scores only · no trust vote



Evidence balance
Distinct study support
minus contradictions



Research knowledge graph
Eligible validated edges
Provisional & audit views stay separate



Researcher findings

PDF preparation and sentence references

Purpose: Make the paper readable and keep later claims tied to their original evidence.

WHAT HAPPENS

Read pages, sections, bibliography entries and citation markers from the PDF.

Number the source sentences and retain neighbouring text for references such as “these grains”.

OUTPUT

Source passages, sentence references and linked bibliography entries.

ILLUSTRATIVE SENTENCE NUMBERING

S12 Fusarium head blight affects wheat grains.

S13 These infected grains may contain mycotoxins.

S12 identifies the grains referred to in S13.

Atomic claims with their source evidence

Purpose: Turn source statements into triplets while preserving conditions and certainty.

SOURCE PAPER

"FHB does cause important yield reductions, but the disease has become a primary concern because of the mycotoxins that may be produced in infected grains"

Crop health and its global impacts on the components of food security, PDF page 6



SUBJECT

Infected wheat grains

RELATION / OBJECT

May contain / mycotoxins

RETAINED CONDITION

Fusarium head blight infection

Output: a claim and exact sentence references.

Validation against the original PDF

Purpose: Check that the source supports the claim as worded before assembling pathways.

WHAT HAPPENS

Compare the entities, relation direction, conditions and certainty with the original passage.

Keep a source mismatch separate from whether other research supports the scientific relationship.

OUTPUT

Source-entailed, unsupported or needs review, with a reason.

EXAMPLE: PRESERVING UNCERTAINTY

Source wording: “may contain”

✓ **Claim: “may contain”**

Preserves the source’s uncertainty.

✗ **Claim: “always contain”**

Adds certainty the passage did not provide.

Shared entities connect provisional pathways

Purpose: Connect source-grounded edges through shared entities before external assessment.

MiMo Flash aligns entities across saved papers; the code connects the claim edges.



SAVED CASE-STUDY STATUSES AFTER EXTERNAL ASSESSMENT

Claim 1: independent supporting evidence

Claim 2: source-grounded; external review required

Related claims are grouped for assessment; every edge keeps its own evidence status.

Literature queries and paper discovery

Purpose: Find studies that can test the claim, including the papers cited by its source.

WHAT HAPPENS

MiMo Flash translates the claim and its conditions into complementary literature searches.

Search Europe PMC and OpenAlex. Also resolve references attached to the original source passage.

OUTPUT

Candidate article metadata, with cited and independent origins kept separate.

EXAMPLE: FINDING MYCOTOXIN STUDIES

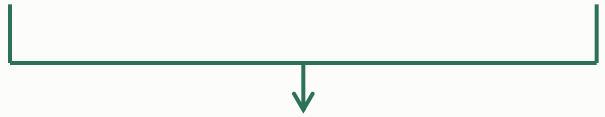
wheat grain + Fusarium + mycotoxins

Independent search

Europe PMC
OpenAlex

Source references

Bibliography
DOI / Crossref



Deduplicated candidate papers

Article ranking and source retrieval

Purpose: Prioritize promising papers and obtain the source text needed to assess them.

WHAT HAPPENS

Use article titles, available abstracts and source links to rank relevance to the claim.

Fetch available full text or an indexed abstract. Keep the evidence level and unavailable sources visible.

OUTPUT

Selected papers with source text or an explicit pending/unavailable status.

HOW A PAPER REACHES THE READER



BGE + BM25

Semantic similarity and keyword relevance

MedCPT cross-encoder

Score claim and article text together

Fetch source text

XML, public PDF or indexed abstract

A high-ranked title is not scientific proof.

Passage preparation and selection

Purpose: Select the parts of each paper that address the claim without losing its conditions.

WHAT HAPPENS

Prepare identified chunks with source sentences and enough context to interpret them.

BGE embeddings find candidates. MedCPT reranks them using the full claim, qualifiers and a neutral endpoint query.

OUTPUT

Selected literal passages with paper, chunk and sentence identifiers.

EXAMPLE OF A PAPER EVIDENCE PACKET



Paper A · identified source text

S12	Crop and disease context
S13	Finding about grain mycotoxins
S14	Study conditions or limitations

Select evidence together with its relevant context.

Per-paper evidence judgement

Purpose: Ask whether one paper supports, contradicts or does not address each related claim.

WHAT HAPPENS

DeepSeek Flash reads selected passages from one paper against the related claims.

Judge the endpoints, relation, conditions and certainty separately for every claim. Return the selected evidence.

OUTPUT

Each claim’s verdict, paper finding and selected proof, with reasons for review.

ILLUSTRATIVE SHARED-PAPER ASSESSMENT



Paper A + claims C1 and C2



DeepSeek Flash assessment

C1	Support with its own evidence
C2	Insufficient evidence in this paper

Evidence traceability and NLI diagnostics

Purpose: Attach findings to their actual source and add a language-consistency diagnostic.

WHAT HAPPENS

Code checks paper, chunk and sentence identities, then reconstructs the exact source quotations.

DeBERTa NLI adds entailment, neutral or contradiction scores for the admitted evidence and the fixed claim.

OUTPUT

Traceable proof and diagnostic scores. The LLM retains the scientific judgement.

EXAMPLE: KEEPING A QUOTE WITH ITS PAPER

✓ **Paper A → Paper A**

Sentence IDs and quote belong together.

✗ **Paper A → Paper B**

A quote cannot move to another paper.

Natural Language Inference (NLI)
Entailment / neutral / contradiction

Cross-paper synthesis

Purpose: Reconcile paper findings and assess the compatibility of neighbouring claim edges.

WHAT HAPPENS

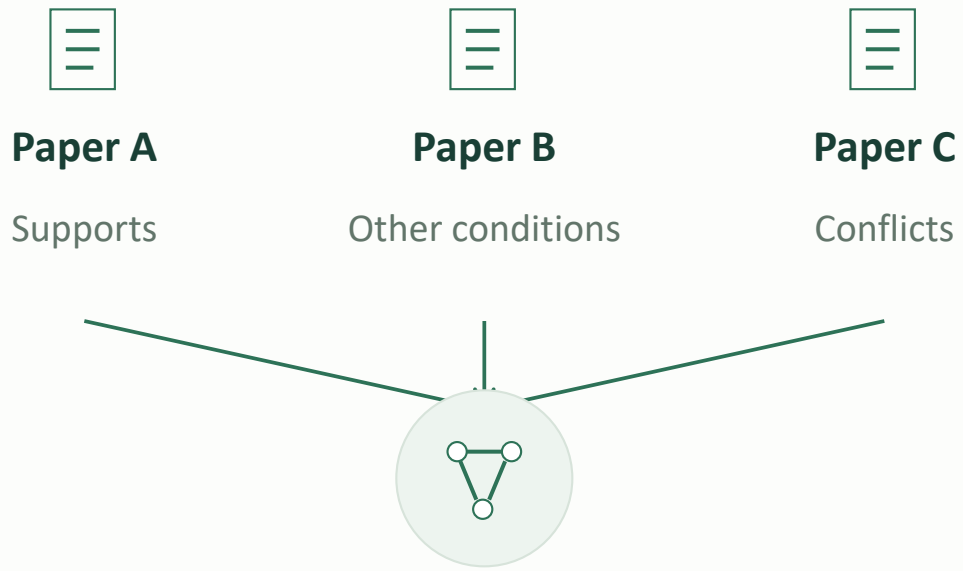
MiMo Pro reads the admitted findings together, with each paper's evidence and conditions.

Reconcile support and conflict for each edge. Check project scope, contextual entity roles and compatible adjacent links.

OUTPUT

Evidence decisions for each claim and a separate assessment of connection compatibility.

ILLUSTRATIVE CROSS-PAPER COMPARISON



Every edge keeps its own conclusion.

Trust evidence and the final graph

Purpose: Record distinct study evidence per edge and select eligible edges for the final graph.

WHEAT-GRAIN / MYCOTOXIN CLAIM

+3

Recorded evidence balance

3

Supporting study groups

0

Contradicting study groups

4

Unresolved paper findings

Evidence balance is a count, not a probability. Graph connectivity is reported separately.

One original study contributes once

Repeated papers, passages or pathways do not add new votes.

Final graph eligibility

Keep source-grounded, in-scope edges with completed assessments that meet evidence requirements.

Saved results for the Crop Health case study

A single-document research snapshot with separate source and external evidence outcomes.

49

Extracted claims

46

Source-grounded claims

8

Retained graph edges

15

Entities in the external graph

External outcome	Claims
Independent supporting evidence	8
Source-reference support	1
Review required	26
Outside project scope	7
Contradiction / contested	2
External evidence not found	1
Retrieval pending	1

46 source-grounded claims enter evidence assessment.
45 have completed assessments; 1 has pending retrieval.

Researchers can inspect a pathway and its evidence

The interface keeps extracted claims, provisional connections and validated edges distinguishable.

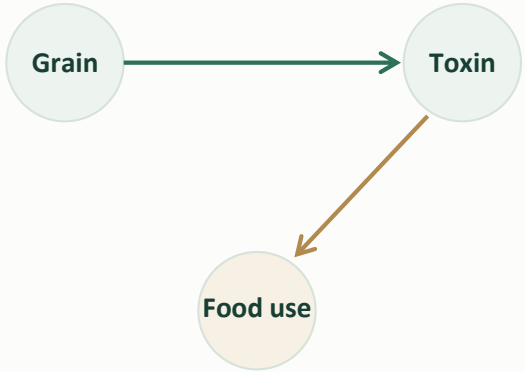
Claim inspector

Original triplet

Source passage and page

External paper findings

Trust network



Evidence balance + review states

Validated graph

8 retained edges

15 aligned entities

Traceable source citations

A connected research graph with an inspectable evidence record for every claim